Nonsinusoidal phase modulations for high-power laser performance control: stimulated Brillouin scattering and FM-to-AM conversion

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High-power lasers, such as the Laser MegaJoule (LMJ), have to be phase modulated to avoid stimulated Brillouin scattering (SBS) that may strongly damage optics at the end of the laser chain. Current spectral broadening on LMJ is performed with a sinusoidal phase modulation. This pure sinusoidal phase modulation leads to inhomogeneous spectral power densities (SPD). Thus, for a same SBS power threshold, the sinusoidal phase-modulated spectrum has to be larger than the equivalent ideal SPD with isoenergetic peaks. We present in this paper a technique to generate energy-balanced Dirac peaks spectra thanks to nonsinusoidal phase modulations. Thus, we can build a narrower spectrum with a nonsinusoidal phase modulation that has the same SBS threshold as a sinusoidal phase modulation, and we show that FM-to-AM conversion can be strongly reduced, which is of great interest for LMJ laser performance, with reductions up to 40%. © 2010 Optical Society of America

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1. Introduction

The Laser MegaJoule (LMJ) in France and the National Ignition Facility (NIF) in the United States are high-power laser facilities designed to achieve inertial confinement fusion of a deuterium tritium (D-T) capsule [1,2]. Such energetic lasers (up to 2 MJ in a few nanoseconds) need broadened spectra for two reasons: to avoid stimulated Brillouin scattering (SBS) and to "smooth" the final focal spot that reaches the D-T capsule. First, without spectral broadening, spectral power density (SPD) of ignition laser pulses would exceed the transverse Brillouin threshold in optics at the end of the amplification chain. Such SBS would strongly damage these optics [3]. Spectral broadening spreads energy to different wavelengths (frequencies) to reduce the SPD maximum level below the SBS threshold. Second, focal spots on the LMJ require a high spatial homogeneity. As a matter of fact, the focal spot of the laser beam is shaped by a phase plate and looks like a speckle pattern with hot spots. The intensity is time averaged through a spectral broadening and a dispersion system (grating) that moves the hot spots during the pulse. This scheme is called smoothing by spectral dispersion [4].

The required spectral broadening on LMJ is generated by a phase modulation [i.e., frequency modulation (FM)] in the fibered part of the front end. Detrimental consequences of phase modulation appear during propagation through the laser chain. Ideally, a pure phase modulation does not change beam intensity. However, propagation through the different components slightly filters the optical

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spectrum. The consequences of these filters are an energy loss and a partial conversion of FM into intensity modulation [i.e., amplitude modulation (AM)]: this is FM-to-AM conversion. For fusion experiments and laser–plasma interaction, the control of the temporal shape is a key point and, thus, FM-to-AM conversion needs to be reduced [5].

In this paper, we only consider the phase modulation used to avoid SBS (the so-called anti-Brillouin function). However, the techniques mentioned in the following can as well be used to perform the smoothing function. The examples of sinusoidal phase modulation presented in Fig. 1 show that the optical spectrum is made of regularly spaced Dirac peaks but energies in these peaks are not equal, except for one particular case with only three peaks [Fig. 1(a)]. Thus, to control the SPD maximum to avoid reaching the SBS threshold, we need to dramatically enlarge the spectrum. Total bandwidth is considerably larger than an equivalent homogeneous distribution of energy and this is a drawback, considering FM-to-AM conversion.

In the following, we describe a method to generate more homogeneous spectral power distributions using nonsinusoidal phase modulations (Section 2). Then, we focus on the anti-Brillouin function and FM-to-AM conversion. We show that we can get a narrower spectrum compared to sinusoidal phase modulation that can achieve the same anti-Brillouin performance on LMJ (Section 3). Also, we experimentally observe that the impact on FM-to-AM conversion is significant on amplitude filters representative of propagation filtering in the LMJ chain (Section 4).

2. Principle, Optimization and Experimental results of Nonsinusoidal Phase Modulation

In this section, we show limitations of sinusoidal phase modulations and we propose to use nonsinusoidal phase modulations to satisfy the same anti-Brillouin requirements on LMJ. First, we justify the current parameters of the sinusoidal phase modulation used to avoid SBS on LMJ. Second, we present the principles of nonsinusoidal phase modulations and a strategy for a user-friendly implementation. Finally, we measure anti-Brillouin function performance to compare both phase modulations.

A. Sinusoidal Phase Modulation

On LMJ, the anti-Brillouin function is performed thanks to a sinusoidal phase modulation that is characterized by two parameters: the frequency f_m and the modulation index m. The optical field is given by

$$a(t) = \sqrt{I(t)} \cdot \exp(i[2\pi f_0 t + m \sin(2\pi f_m t)]), \qquad (1)$$

where $I(t)$ is the pulse intensity, and f_0 is the optical frequency of the laser. The contribution of the phase modulation, a_{mod}, can be expressed by

$$a_{\mathrm{mod}}(t) = \exp(im \sin(2\pi f_m t)). \qquad (2)$$

So the optical spectrum of the phase-modulated signal is the convolution of the initial temporal shape spectrum and the phase modulation spectrum.

On LMJ, the temporal pulse shape bandwidth is about 0.5 GHz (>95% of the energy). Thus, in the following, the optical spectrum is given in the base band and is deduced from Eq. (2). $a_{\mathrm{mod}}(t)$ is a periodic signal and the corresponding optical spectrum is composed of uniformly separated f_m spaced Dirac peaks:

$$\tilde{a}_{\mathrm{mod}}(f) = \sum_{n=-\infty}^{+\infty} J_n(m)\delta(f - nf_m), \qquad (3)$$

where $J_n(m)$ is the nth Bessel function [6].

Phase modulation spreads the pulse energy on different wavelengths. Energy in the spectral domain is given by the spectral power distribution defined as

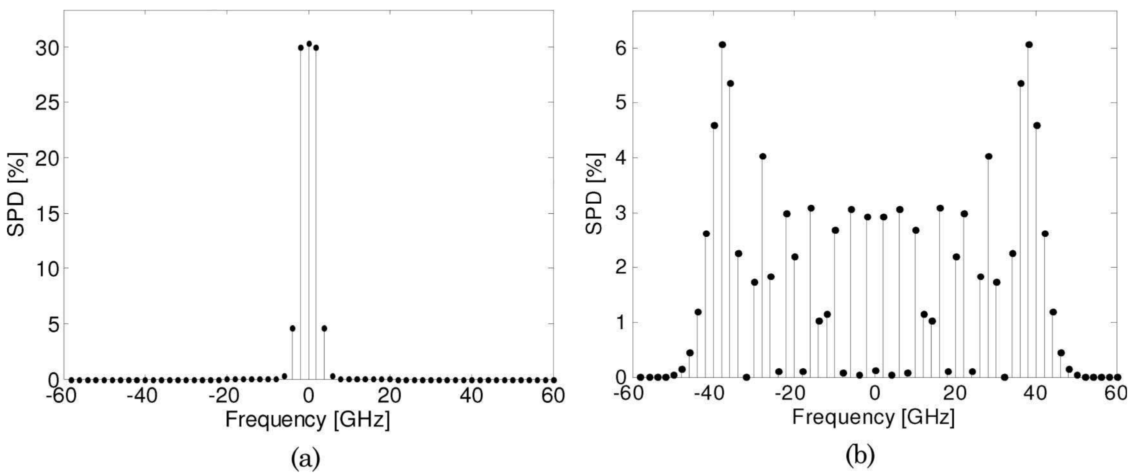


Fig. 1. Different optical spectra obtained by sinusoidal phase modulation. (a) Energy-balanced spectra with three peaks for $m = 1.43$ rad ($f_m = 2$ GHz). (b) Normalized spectral power of density of LMJ anti-Brillouin function sinusoidal phase modulation at 3ω: $f_m = 2$ GHz and $m = 21$ rad.

$$\mathrm{SPD}(f) = |\tilde{a}_{\mathrm{mod}}(f)|^2 = \sum_{n=-\infty}^{+\infty} |J_n(m)|^2 \delta(f - nf_m). \quad (4)$$

Without phase modulation, 100% of the energy is in the central frequency. With a sinusoidal phase modulation, energy in each peak is thus given by the functions $|J_n(m)|^2$. These functions are odd for the variable n and. thus, for any sinusoidal phase modulation, the spectrum barycenter remains the laser carrier frequency and, thus, no change of the carrier frequency is required when phase modulation is set on and off.

Note that, on LMJ, amplification of the laser signal is generated in the IR domain (1053 nm or 1ω) but laser–target interaction is made in the UV (351 nm or 3ω). Simple considerations on the frequency conversion system show that the optical field $a_{3\omega}$ at 3ω is linked to the optical field $a_{1\omega}$ at 1ω and $a_{3\omega}(t) \propto a_{1\omega}{}^3(t)$. Considering Eq. (2), one can see that phase modulation remains sinusoidal with frequency conversion, but the modulation index at 3ω is $m_{3\omega} = 3m_{1\omega}$ [7].

B. Anti-Brillouin Requirements

Considering the anti-Brillouin function, SBS can be suppressed in high-power lasers if energy is spread on different wavelengths for the photons to excite independently different Stokes modes. This condition is satisfied if the frequency difference between Dirac peaks is larger than the considered FWHM Brillouin linewidth of gain spectrum $\Delta\nu_B$. For silica at 351 nm, the transverse Brillouin linewidth is $\Delta\nu_B \sim 200$ MHz [8]. Thus, broadening is efficient if SPD peaks are separated by more than $\Delta\nu_B$. On LMJ, the anti-Brillouin modulation frequency has been chosen at $f_m = 2$ GHz (on NIF, $f_m = 3$ GHz).

For the LMJ frequency modulation, each of the SPD frequency peaks excites one different Stokes mode. The only condition to avoid SBS is, then, that all these SPD peaks have to be below the SBS power threshold. The current anti-Brillouin function has a modulation index at 3ω $m_{3\omega} = 21$ rad. Therefore, in the following, we consider that the anti-Brillouin function is performed for 2 GHz phase-modulated signals if the SPD's most energetic peak is less than 6% of the total energy at 3ω, as in the sinusoidal modulation case [presented in Fig. 1(b)], by taking to account the LMJ requirement.

C. Nonsinusoidal Phase Modulations: Interest and Choices

As mentioned in Eq. 3 and the examples in Fig. 1, SPD peaks evolve with Bessel functions and do not carry the same energy for sinusoidal phase modulations. It is not optimal if we want to minimize spectral broadening. In the following, we consider the spectrum bandwidth as the frequency range corresponding to 95% of the total energy. For sinusoidal phase modulation, the bandwidth, Δf, is given by the Carson rule that links the bandwidth to the phase modulation parameters m and f_m: $\Delta f = 2 \cdot (m+1) \cdot f_m$ [9]. For example, at 3ω, LMJ anti-Brillouin function broadening is 88 GHz. Note that the ideal spectrum to achieve an anti-Brillouin function is a perfectly energy-balanced SPD, which means with the same energy in every peak. Thus, with only 17 peaks of less than 6% of the total energy, we have the same anti-Brillouin performance with a broadening of only 32 GHz, as shown in Fig. 2. However, this ideal spectrum cannot be obtained with a pure phase modulation, as shown in Appendix A. We are looking for solutions close to this ideal case and we want another phase modulation that keeps the anti-Brillouin function with a narrower spectral bandwidth. The 2 GHz spaced peaks requirement is still valid, so we are looking for a 500 ps periodic signal. As the signal is periodic, it can be developed in Fourier series as

$$a_{\mathrm{opt}}(t) = \exp\left(i \sum_{n=1}^{\infty} m_n \sin(2\pi n f_m t + \varphi_n)\right), \qquad (5)$$

with m_n as the different modulation indexes and φ_n as the related phases.

However, in practice, the number of harmonics is limited by phase modulator bandwidth (about 40 GHz). Therefore, the number of harmonics is limited to N:

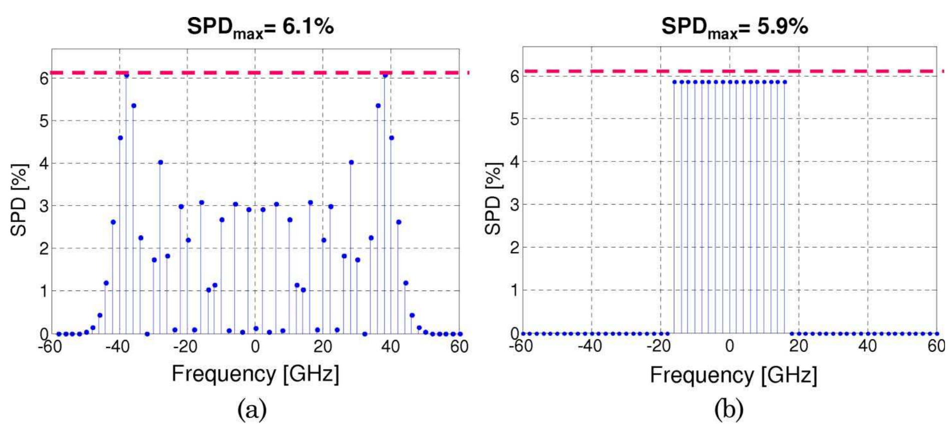


Fig. 2. (Color online) SPD to achieve anti-Brillouin function. (a) Sine phase modulation SPD at 3ω ($m = 21, f_m = 2$ GHz). (b) Ideal energy-balanced spectrum with same value of maximum of SPD (about 6% of the total energy).

$$a_{\mathrm{opt}}(t) = \exp\left(i \sum_{n=1}^{N} m_n \sin(2\pi n f_m t + \varphi_n)\right). \quad (6)$$

The limited value of N prevents us from reaching perfectly energy-balanced spectra. However, nonsinusoidal phase modulation [or multisinusoidal-phase-locked phase modulation considering Eq. (6)] can provide better spectral shapes than sinusoidal phase modulation. In the following, we study spectral performance for different values of N.

Phase values φ_n are important. Indeed, Eq. (6) shows that φ_n values determine how the spectra from each harmonic interferes. In particular, nonsinusoidal phase-modulation SPDs may not be symmetrical and barycenter may not be the carrier frequency. Note that we can set $\varphi_1 = 0$ as a time reference choice. Then, the number of freedom degrees is equal to $2N - 1$.

To quantify the spectrum quality, we simply consider an error function $\varepsilon(\mathrm{SPD})$ that is the norm of the difference between the nonsinusoidal phase modulation SPD and an ideal energy-balanced spectrum ($\mathrm{SPD}_{\mathrm{ideal}}$):

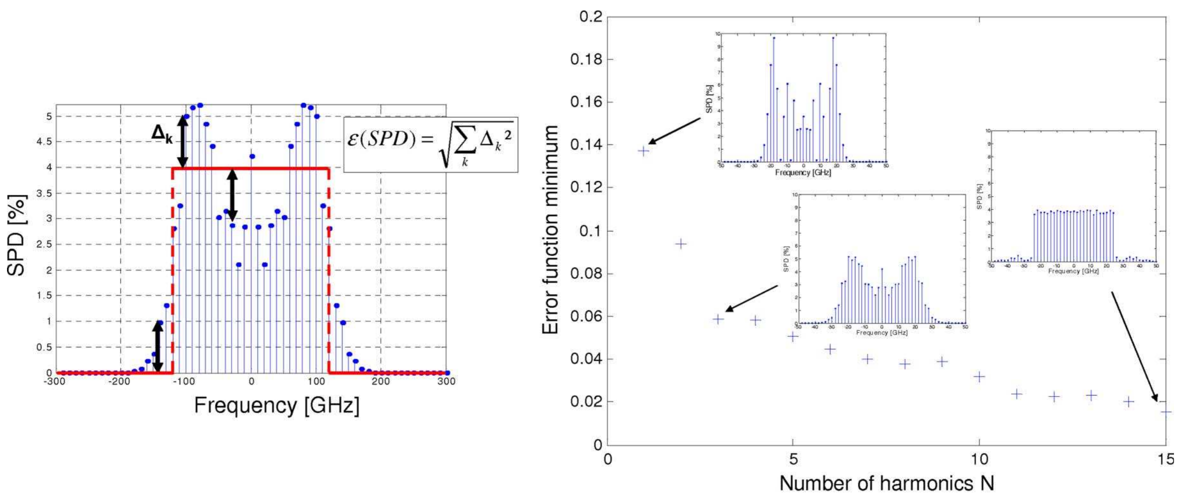
$$\varepsilon(\mathrm{SPD}) = \sqrt{\sum_{k=-\infty}^{+\infty} (\mathrm{SPD}(kf_m) - \mathrm{SPD}_{\mathrm{ideal}}(kf_m))^2}. \quad (7)$$

The definition used in Eq. (7) is illustrated in Fig. 3(a).

Optimization is performed thanks to a gradient algorithm with a random starting point in the $(2N - 1)$ dimension solution space. For each value of N, the algorithm has been running more than 100,000 times. An adaptation of the Gerchberg–Saxton algorithm has also been tested: for each iteration, the temporal signal is filtered to keep only the N first harmonics [10]. It is faster (a few minutes instead of hours), but error function values remain higher than with the gradient algorithm. Figure 3(b) represents error function optimization for different values of the number of harmonics N, considering an arbitrary ideal spectrum of 25 peaks. For $N = 15$, the best spectrum is very close to the ideal energy-balanced spectrum. However, one can see that important gain on the error function is already obtained for $N = 3$, for which cost and complexity are still low (five parameters). Furthermore, if we want to use an equivalent system to smooth the beam (i.e., FM $f_m = 14.25$ GHz), only the first three harmonics are transmitted with a commercially available 40 GHz phase modulator.

Among all the spectra obtained with nonsinusoidal phase modulations, symmetrical SPDs are particularly interesting. Indeed, they avoid shifting the laser carrier frequency, considering that the amplifier laser cavities are set for a particular wavelength. For $N = 2$, SPD is an odd function if $\varphi_2 = \pi/2\ [\pi]$. In the case of $N = 3$, symmetrical positions are limited by the phase choices and $\varphi_2 = \pi/2\ [\pi]$ and $\varphi_3 = 0\ [\pi]$. Demonstration of these assertions is given in Appendix B. Hopefully, these phase choices for symmetrical SPDs also correspond to minimum values of $\varepsilon(\mathrm{SPD})$, as shown in Fig. 4 for the particular choice of a spectrum with 25 peaks.

Nonsinusoidal phase modulation use for the anti-Brillouin function has been studied before, but it only appears in [11,12]. Both of them are related to the particular operating point of the sinusoidal phase modulation when $m \approx 1.4$ rad. As presented in Fig. 1(a), SPD for $m = 1.43$ rad is made of three equal peaks. The principle in [11,12] is to add a second phase modulation at the third harmonic with the same value of m: convolution of spectra leads to a spectrum made of nine energy-balanced peaks. This



(a) (b)

Fig. 3. (Color online) (a) Illustration for the definition of the error function ε used when we are looking for an ideal 25-peak energy-balanced spectrum. (b) Results of gradient algorithm optimization for nonsinusoidal phase modulation for different values of the number of harmonics: the higher N is, the lower the error function is. However, most of the gain is obtained for $N = 3$, which is simple enough for practical realization.

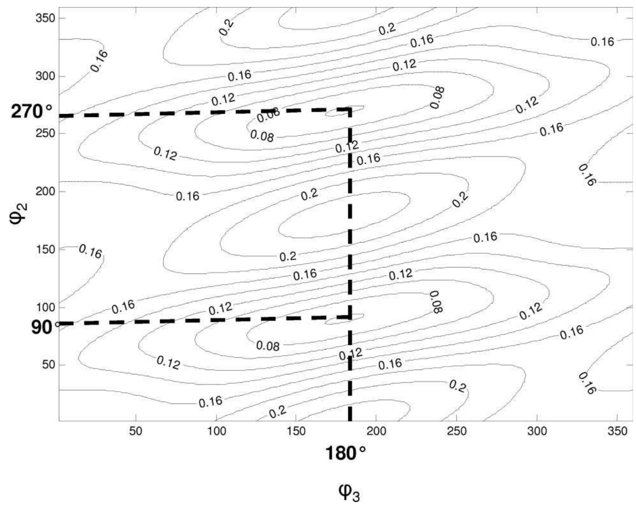


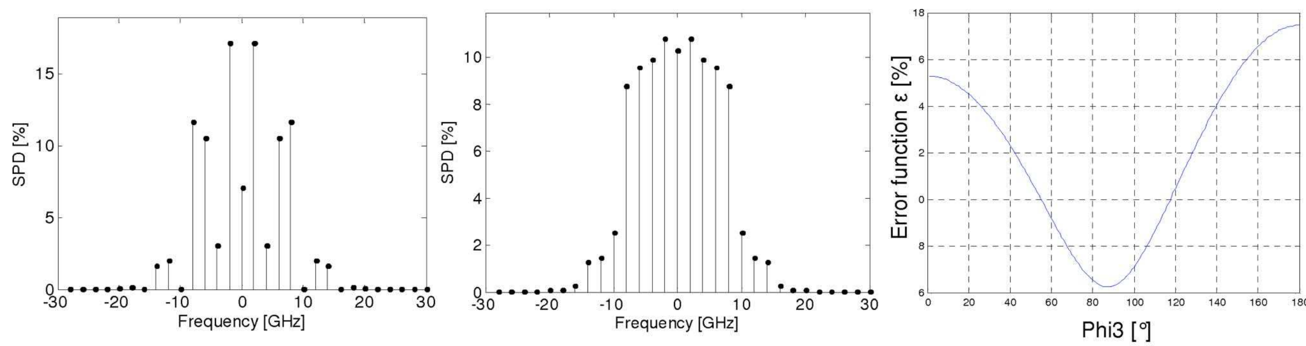
Fig. 4. Error function for nonsinusoidal phase modulation with $N = 3$ for any values of phases φ_2 and φ_3. In that example, the ideal spectrum considered is a 25-peak spectrum and $m_1 = 1.4$, $m_2 = 5.6$, and $m_3 = 0.4$. Minima of error function are symmetry positions for SPD.

idea is generalized to spectra composed of any power of three peaks. However, as it appears in Fig. 5(a), the energy is not totally concentrated in the three central peaks. The impact of the remaining energy (about 10%) in the other peaks is important and limits the performance of such systems. Furthermore, [11,12] do not consider any phase choice between the harmonics, which leads to systems that do not work and are judged by the authors as not stable, which is logical with a lack of phase control and as presented in Fig. 6. Indeed, [11] uses different frequency oscillators to generate the harmonics and [12] uses one generator, but does not control the relative phases that are affected by amplifiers, attenuators, low-pass filters. or time delay before recombination. Phase control is the key point of our method, as well as the choice of any number of peaks and not only the power of three.

For optimized solutions, values of m_1 and m_3 are almost constant for spectra with a bandwidth between 10 and 40 peaks (larger spectra have not been studied). These values are $m_1 = 1.4 \pm 0.1$ and $m_3 = 0.4 \pm 0.1$ and, to set the spectrum bandwidth, only m_2 has to be modified, as illustrated in Fig. 6. It is interesting, since it means that, even with a more complex signal to generate the phase modulation, all the parameters except one are fixed and the last parameter (m_2) set the spectrum bandwidth: optimized nonsinusoidal phase modulation with a $N = 3$ tuneable bandwidth becomes as simple as sinusoidal phase modulation to set [13].

Recently, another team has published results on the same topic [14]. Since they want to avoid SBS in fiber, the frequency modulation used ($f_m = 50\,\mathrm{MHz}$) is much lower than our frequency modulations (2 and 14.25 GHz). So they are not limited by the frequency modulator bandwidth and the cost of components remains low and the use of many harmonics remains possible. However, the drawback is the $2N - 1$ degrees of freedom that might be hard to control, instead of our solution with only one degree of freedom. In their solutions, they use only odd harmonics, where the optical spectrum remains symmetrical whatever the relative phase. However, to get efficient spectra, relative phase relations close to multiples of $\pi/2$ are necessary. However, they have other constraints on modulation indices that we do not have: they consider $m < 4$ for each harmonic, whereas we can use our modulators to reach $m > 10$, keeping a linear response. Thus, the proposed solutions are different.

In the next section, we present experimental results for nonsinusoidal phase modulation. We show that, with our system, we can perform better energy-balanced spectra and we measure the SBS threshold in particular, through representative configurations, and we show that anti-Brillouin function can be performed with a narrower spectrum with our system.



(a) (b) (c)

Fig. 5. (Color online) Importance of phase control. Particular operating point for a nonsinusoidal phase modulation composed of two harmonics: $f_m = 2\,\mathrm{GHz}$ and $3f_m = 6\,\mathrm{GHz}$ with the same modulation index $m = 1.4$. Two U.S. patents [11,12] assume that the SPD associated is composed of nine energy-balanced peaks. However, (a) if $\varphi_3 = 0$, as presented in patent from [12], SPD is not energy balanced. (b) Energy-balanced spectrum is performed only for a good choice of the relative phase: $\varphi_3 = \pi/2\ [\pi]$. (c) Evolution of error function in the case of an ideal energy-balanced spectrum with nine peaks for different values of φ_3.

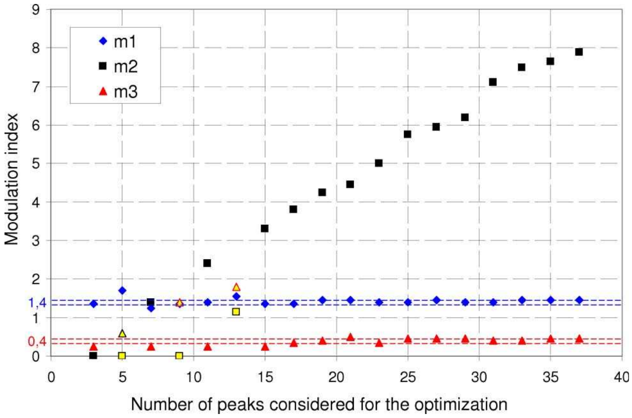


Fig. 6. (Color online) Modulation indexes for optimized three-harmonic-signal solutions. (m_1, m_3) are almost constant and m_2 increases with the spectrum width. Exceptions are presented in contrasting shades.

D. Experimental Results

For experimental purpose, we have built a practical system able to generate the first three harmonics from a 2 GHz single oscillator. We present the system configuration in Fig. 7. Controlling the relative phases between harmonics at these frequencies is easy because a 2π phase shift corresponds to 100 ps at 10 GHz, i.e., 1–2 cm in a coax cable. We decided to combine the harmonics in the high frequency signal part and so use only one phase modulator. Other configurations can be developed and we could imagine a system with a phase modulator for each harmonic.

Figure 8 shows different spectra measured with a Fabry–Perot interferometer that can scan the optical spectrum domain with a free spectral range of 40 GHz (equivalent to a maximum of twenty 2 GHz spaced peaks; in practice, we consider spectra with a maximum of 15 peaks to avoid aliasing): for any of these spectra, m_1 and m_3, as well as φ_2 and φ_3, are set as presented in Subsection 2.C: there is a good agreement between measured spectra and numerically simulated spectra. Energy in peaks is better balanced for nonsinusoidal than for sinusoidal phase modulators.

In the following, we check that, because energy is better spread among the optical spectrum, the anti-Brillouin function can be performed with a narrower spectrum by comparing SBS thresholds for sinusoidal and nonsinusoidal phase modulations. For practical reasons, we choose to measure the SBS threshold for backscattering in fiber at 1053 nm. The only change, considering the SBS parameter, are the Brillouin linewidth and the power threshold: the Brillouin linewidth for backscattering in bulk silica at 1053 nm is $\Delta\upsilon_B \sim 50$ MHz and is confirmed in optical fiber [8]. Thus, the choice of $f_m = 2$ GHz between each peak will stimulate different Stokes. We measure the required injected power, P_{inj}, to get the same stimulated Brillouin backscattering (SBB) level. The spectra are characterized by the maximum value of their spectral power SPD, $\text{SPD}_{\max}$, expressed in fraction of the total power: $\text{SPD}_{\max} = 1$ for a nonbroadened spectra and decreases with enlargement. Then, if each peak stimulates a different Stokes mode, P_{inj} follows the rule

$$P_{\text{inj}} = \frac{P_{\text{Threshold}}}{\text{SPD}_{\max}}, \tag{8}$$

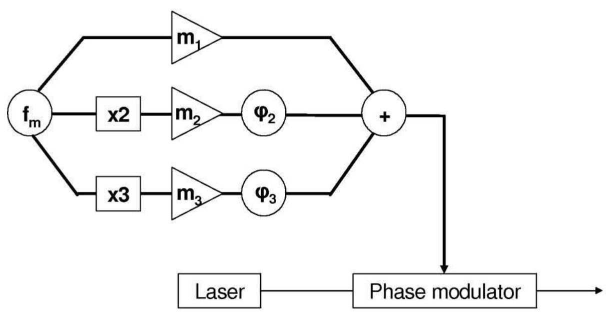
with $P_{\text{Threshold}}$ as the required power to reach the SBS threshold level without broadening. And so P_{inj} increases when $\text{SPD}_{\max}$ decreases for a given $P_{\text{Threshold}}$ value.

As presented in Fig. 9(a), energy-balanced spectra reduce $\text{SPD}_{\max}$ values compared to sine phase modulation spectra for a given broadening. So, for a given threshold, one can use narrower nonsinusoidal phase modulation spectra instead of sinusoidal phase modulation.

We verified Eq. (8) experimentally. For a single-mode fiber, evaluation of the power threshold is given by [15]

$$kg_b P_{\text{Threshold}} L_{\text{eff}}/A_{\text{eff}} \approx 21, \tag{9}$$

with g_b as the peak value of the Brillouin gain. For bulk silica, g_b is about 5×10^{11} m/W. If we consider a



(a)

(b)

Fig. 7. (Color online) (a) Nonsinusoidal phase modulation scheme: the harmonic combination and phase control is performed directly with the HF signal. Optical components remain the same. (b) HF multiharmonic generator at 2–4–6 GHz used in our different setups.

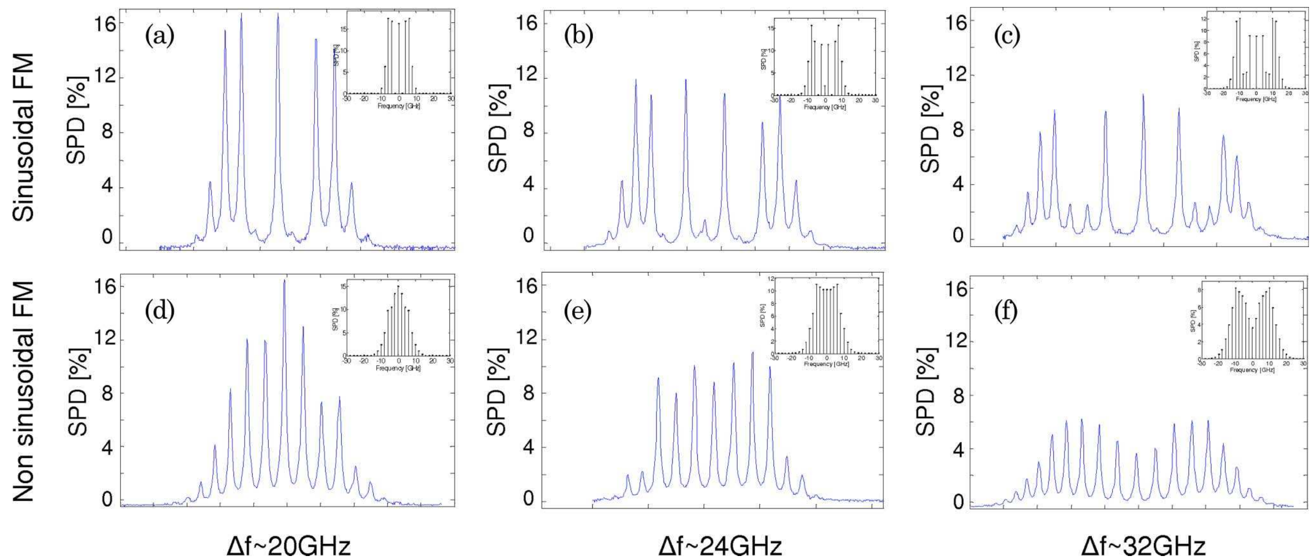
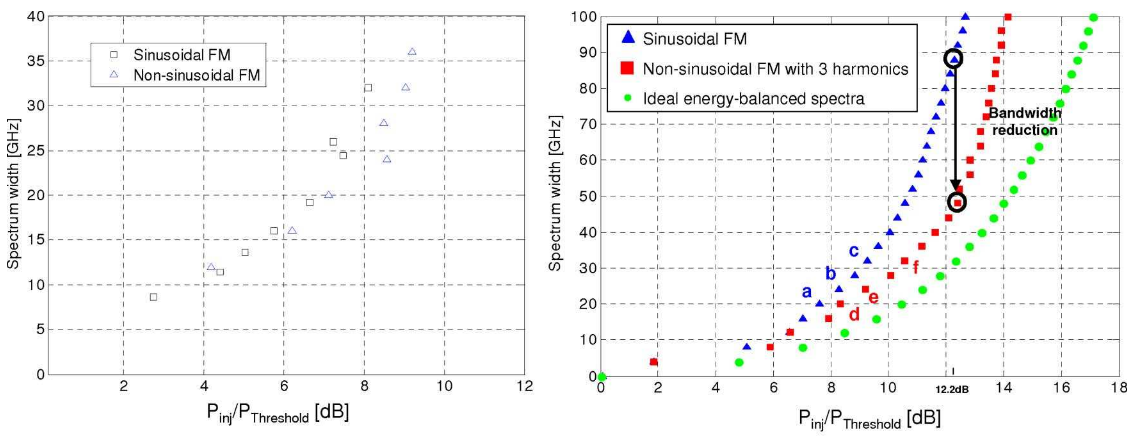


Fig. 8. (Color online) Top line: different nonsinusoidal phase modulation SPD [(a) $m = 2.4$, (b) $m = 5.1$, (c) $m = 7$]. Bottom line: Sinusoidal phase modulation SPD with equivalent bandwidth [(d) $m_2 = 1.8$, (e) $m_2 = 2.05$, (f) $m_2 = 3.3$, $m_1 = 1.4 \pm 0.1$, $m_3 = 0.4 \pm 0.2$, $\varphi_2 = 90°$ and $\varphi_3 = 180°$]. Vertical scale is the same for both lines: nonsinusoidal phase modulation SPD maxima are below equivalent sinusoidal phase modulation SPD maxima.

non polarization maintaining fiber, k is between $1/3$ to $2/3$ [16]. A_{eff} is the effective core area of the fiber ($A_{\text{eff}} = 30\,\mu\text{m}^2$ for a single-mode fiber at 1053 nm) and L_{eff} is the effective fiber length linked to the fiber length and to the linear loss α by [15]

$$L_{\text{eff}} = [1 - \exp(-\alpha L)]/\alpha. \tag{10}$$

As presented in Fig. 10, in our setup $L_{\text{eff}} = 1.6$ km since we used a 2 km optical fiber length with measured loss of 0.9 dB/km. The laser is pulsed but shape controlled: gain of the amplifier is constant to prevent temporal distortion that would modify the SBB signal shape; the power level is set with a variable attenuator before injection into the long fiber. SBB measurements are made with two photodiodes synchronized for both original and back SBB signals: we consider that the SBS threshold is reached when SBB appears over the Rayleigh scattering with always the same level. Criterion on the experimental detection threshold is arbitrary but, because SBS evolves exponentially, any criterion leads to similar power levels. According to Eq. (4), the power threshold should be equal to 11 dBm, close to the experimental value of 12.1 dBm. As shown in Fig. 9(a), when the spectrum is broadened, the maximum of SPD peaks becomes less energetic and therefore, the more powerful the pulse has to be to get back the same SBB level. Thus, influence of energy-balanced spectra is important: narrower spectra reached same anti-Brillouin function performance as sinusoidal phase modulation spectra.



(a) (b)

Fig. 9. (Color online) (a) Experimental measurements. (b) Power P_{inj} required to observe SBS in function of SBS threshold for a not-broadened spectrum using Eq. (8). Discrepancies between (a) experiments and (b) simulations are small and equivalent between sinusoidal and nonsinusoidal cases. The letters (a) to (f) in the figure refer to the spectra in Fig. 8.

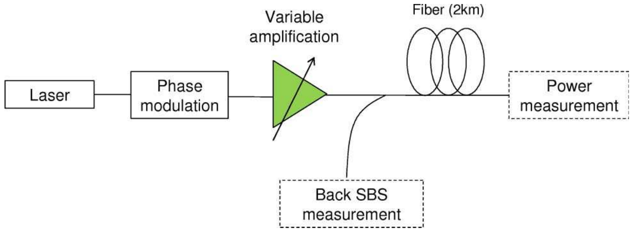


Fig. 10. (Color online) Setup of SBS measurement experiment. Laser is pulsed but shape controlled: gain of the amplifier is constant to prevent temporal deformation; power lever is set with a variable attenuator before injection in the long fiber. The phase modulation used is either sinusoidal or nonsinusoidal. SBS measurements are made with two photodiodes synchronized for both original and back SBS signals: we consider that the SBS threshold is reached when back SBS appears with always the same level.

Figures 8(c) and 8(e) presented two measured spectra that have similar SPD maximum (12%); we verify that the SBS thresholds are about the same in our experiment and we notice that the nonsinusoidal phase modulation spectrum is 25% narrower that the sinusoidal one (24 instead of 32 GHz).

Experimental measurements were limited by broadening capabilities (we use a 1ω design, large spectra as on LMJ at 3ω were not reachable with our setup). Since we agreed that the SBS threshold is directly linked to the maximum of SPD peaks, and we extend this results to larger spectra from Fig. 9(b), where the benefits of nonsinusoidal phase modulation are even more important. In the case of the LMJ anti-Brillouin function at 351 nm, $m = 21$ rad and has a broadening of 88 GHz. An ideal, perfectly energy-balanced spectrum would have the same efficiency for a broadening of 32 GHz. With our nonsinusoidal phase modulation with a variable m_2, we have the same efficiency for a broadening of 48 GHz: same anti-Brillouin efficiency is obtained with a gain of 45% of the spectrum bandwidth.

In the next section, we assess FM-to-AM conversion performance when reducing the bandwidth thanks to nonsinusoidal phase modulations.

3. Reduction of FM-to-AM Conversion with Nonsinusoidal Phase Modulations

Phase modulation does not change the intensity shape. But, if any optical device during propagation changes the optical spectrum (so the optical field by inverse Fourier transform), the intensity will be modulated; this is FM-to-AM conversion. To quantify the FM-to-AM conversion, the distortion criterion α defined by the following expression is used:

$$\alpha = 2 \cdot \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}. \tag{11}$$

α ranges from 0 to 200% and is ideally equal to 0 (no intensity variation).

On LMJ, FM-to-AM conversion is mostly due to two different effects: symmetrical amplitude filters and phase filtering (chromatic dispersion). Amplitude filters that are symmetrical for the central spectrum wavelength have different origins in a LMJ chain: gain narrowing in the preamplifier module, as well as in the amplification section and the limited spectral acceptance of the frequency conversion.

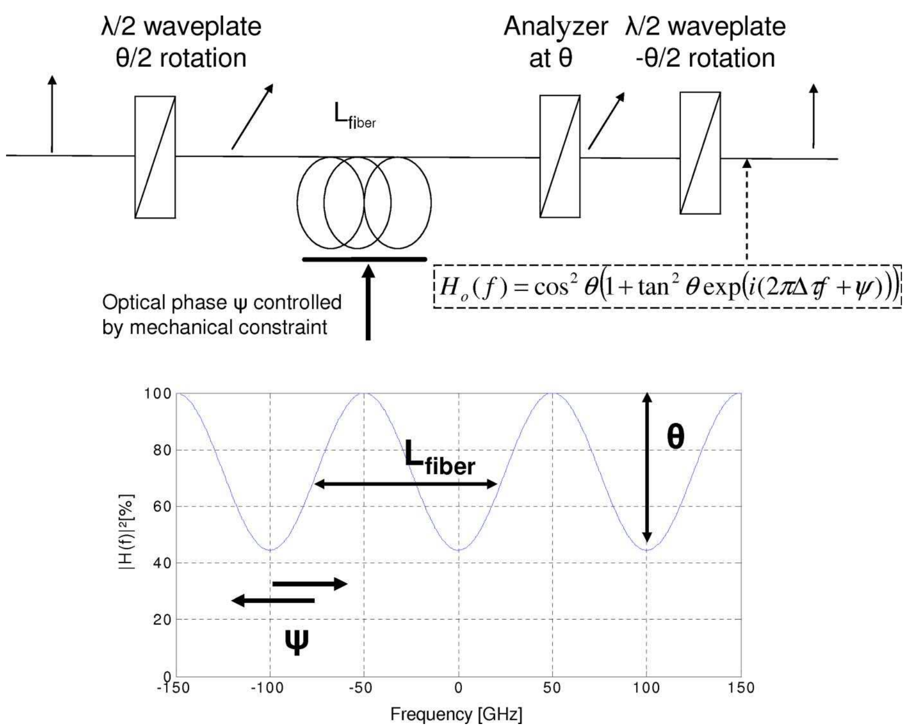


Fig. 11. (Color online) Parametrical optical filter using birefringence in PMF: impacts of the different parameters on the transfer function. The second wave plate is useful to get back the linear polarization on a fiber axis.

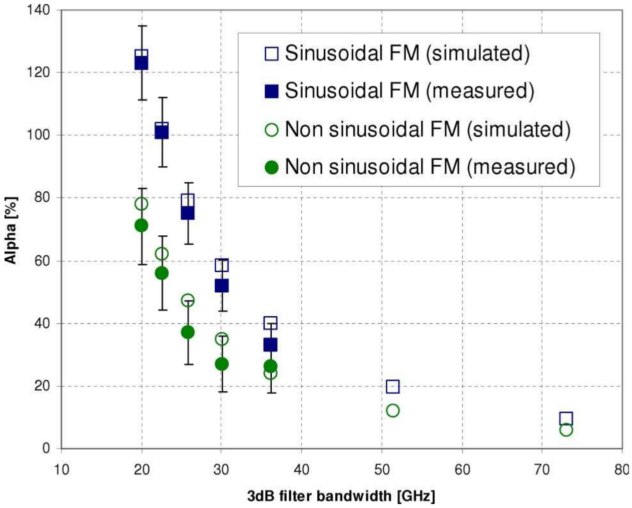


Fig. 12. (Color online) Distortion criterion for different symmetrical amplitude filters. Comparison between sinusoidal (squares) and nonsinusoidal (circles) phase modulation with the same anti-Brillouin efficiency. We added numerical simulations with MATLAB.

Chromatic dispersion has two origins: long fibers in the source (about 200 m) and the propagation between the plane grating and the focusing grating in the frequency conversion system. All these effects have already been studied in [7].

Simple considerations predict that the impact of a phase modulation change is different for these two types of transfer function. Chromatic dispersion study from [7] shows that distortion criterion, α, evolves with $m.{f_m}^2$ for a sinusoidal phase modulation, and so is not directly linked to the broadening $\Delta f = 2(m+1)f_m$. In the case of a nonsinusoidal phase modulation, as m_2 is usually higher than m_1 and m_3 (see Fig. 6) and a first-order $m.{f_m}^2$ becomes $4m_2.{f_m}^2$, the impact of the chromatic dispersion is even more detrimental. Nonsinusoidal phase modulations are not interesting for minimizing the impact of chromatic dispersion. However, compensation of dispersion is possible [17]. On the other hand, analytical results for symmetrical amplitude filters show that the associated distortion criterion evolves with $(m \cdot f_m)^2$ in the case of a sinusoidal phase modulation, which means that $\alpha \propto \Delta f^2$, with Δf as the optical bandwidth. Thus, bandwidth reduction is of interest in minimizing FM-to-AM conversion from symmetrical amplitude filters.

To experimentally check this result, we built a parametrical optical filter as presented in Fig. 11. The transfer function can be rewritten as follows:

$$H_0(f) \propto 1 + A\exp(i(2\pi f\Delta\tau + \Psi)). \qquad (12)$$

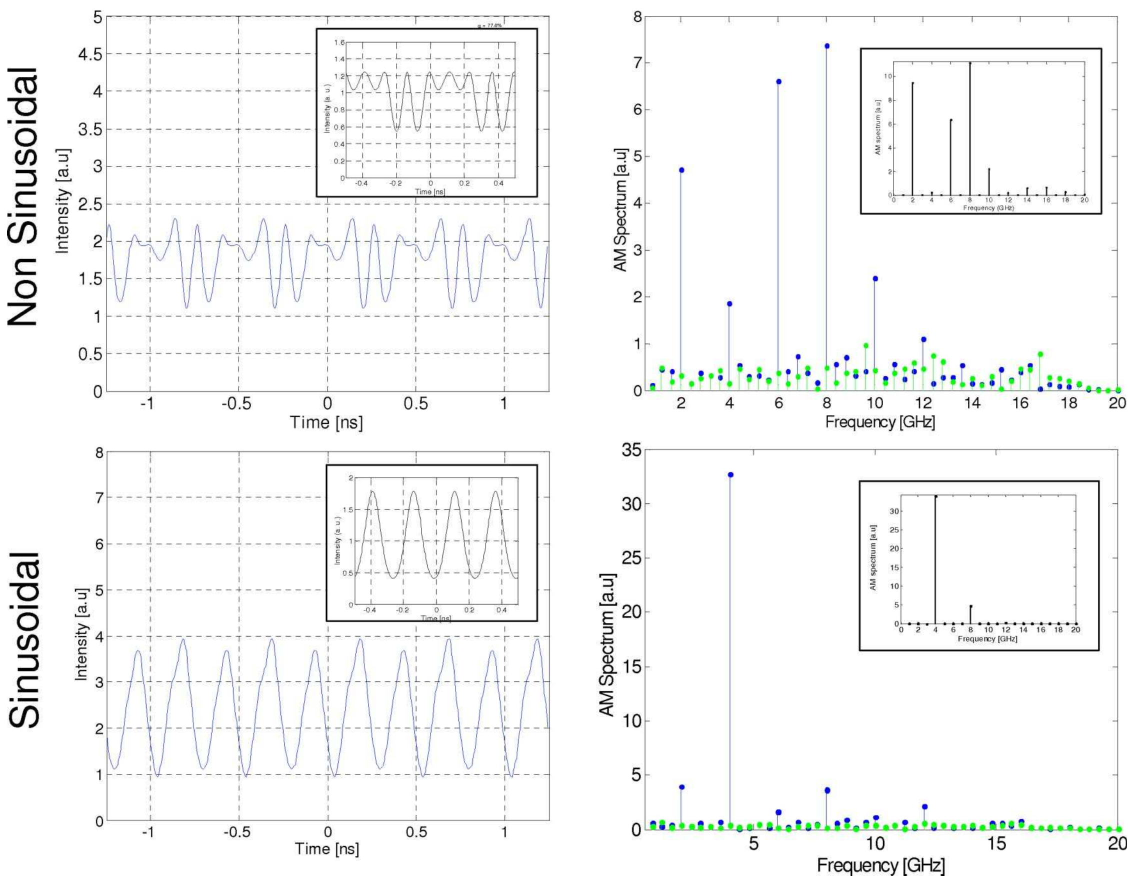


Fig. 13. (Color online) Left column: temporal shape measured after filtering ($f_{3\,\mathrm{dB}} = 20\,\mathrm{GHz}$) for both sinusoidal (bottom line) and nonsinusoidal (upper line) phase modulation. Right column: AM spectra associated. Data points concentrated at the frequency axis represent the measurement noise filtered for the results. In insert, we give numerical simulations obtained with MATLAB.

The interest of this filter is that it can be considered as a symmetrical amplitude filter when we set $\psi = 0°$. Indeed, the phase transfer function is negligible since $H_0(f)$ is then an odd function: the second-order term of the phase transfer function is null. Thus, we may consider that we have the following amplitude transfer function:

$$|H_o(f)|^2 \propto 1 + C\cos(2\pi f \Delta\tau), \tag{13}$$

where $C = 2A/(1+A^2)$ is the contrast, set to 0.9 in our experiment. With this value, the 3 dB cutoff frequency is equal to $f_{3\,\mathrm{dB}} = 1.67/\pi\Delta\tau$. $\Delta\tau$ is proportional to the fiber length, with a measured proportionality coefficient of 1.5 ps/m, and set the filter bandwidth: the longer the fiber, the more severe the filtering. Our experiment was to set this filter at $\psi = 0°$ for different fiber lengths. Then we measured AM modulations for both sinusoidal and nonsinusoidal phase modulation. The measurements were made with a high-speed photodiode and an oscilloscope (12 GHz 3 dB bandwidth). Thus, we use spectra from Figs. 8(c) and 8(e) for comparison since we proved in Section 2 they have same anti-Brillouin efficiency. Results with incertitude estimations are given in Fig. 12 and compared with analytical simulations made with MATLAB. Incertitude origins are multiple, and we considered measurement noise, ψ control, and spectrum control to give the incertitude level. However, experimental results are close to simulations and reduction of FM-to-AM conversion by using nonsinusoidal phase modulation is important: FM-to-AM conversion reduction is better than 40% of total AM modulation.

Another interest of nonsinusoidal phase modulation on FM-to-AM conversion is that the measured temporal shape presented AM modulation at higher frequencies, as it is presented in Fig. 13. Most of AM modulation for sinusoidal phase modulations is included in a $2f_m$ bandwidth. Since nonsinusoidal is a combination of the first three harmonics of f_m and not only f_m, a consequence is that the AM spectrum is more complex and has significant contribution at higher frequencies than $2f_m$. This is interesting because high-frequency AM is less detrimental in a high-power laser. Indeed, frequencies above 7 GHz will by filtered by the focalization to the D-T capsule [7] and, so, FM-to-AM conversion distortion reduction with nonsinusoidal phase modulation will be improved after the final focalization to the target.

4. Conclusions

In high-power lasers, phase modulation is required to avoid SBS that would damage optics. Sinusoidal phase modulation is used, but associated SPD does not have an optimized shape. To minimize enlargement with the same anti-SBS efficiency, homogeneous SPD is an ideal solution. We have shown that it is possible to generate spectra close to this ideal case with a nonsinusoidal phase modulation made of a combination of three harmonics with controlled relative phases. This solution is easy to implement because we have proven that we can have variable broadening spectra by changing only one parameter. Experimental tests are satisfying and we proved that, on LMJ, we can have the same anti-Brillouin efficiency with a 45% narrower spectrum with our solution. Bandwidth reduction is interesting for reducing FM-to-AM conversion. FM-to-AM conversion is an important problem in controlling the temporal shape of pulses. We have shown that, with a nonsinusoidal phase modulation with the same anti-SBS efficiency, the distortion criterion of symmetrical amplitude filters can be reduced up to 40%.

Appendix A

In this appendix, we prove that no pure phase modulation leads to perfectly energy-balanced spectra.

Let us consider a perfectly energy-balanced spectrum with an arbitrary odd number of peaks noted as $2M+1$. Its SPD is then

$$\mathrm{SPD}(f) = \frac{1}{2M+1}\sum_{k=-M}^{M}\delta(f - kf_m). \tag{A1}$$

All the optical spectra, $\tilde{A}$, that have this SPD are

$$\tilde{A}(f) = \frac{1}{\sqrt{2M+1}}\sum_{k=-M}^{M} e^{i\phi_k}\delta(f - kf_m), \tag{A2}$$

where Φ_k are the relative phases between peaks that are degrees of freedom in that proof. The associated optical field is then

$$A(t) = \frac{1}{\sqrt{2M+1}}\sum_{k=-M}^{M}\exp(i(2\pi k f_m t + \phi_k)). \tag{A3}$$

The optical field from Eq. (A3) is a pure phase modulation if $I(t) = |A(t)|^2 = 1$. One can find that

$$\begin{aligned} I(t) = A(t)\cdot A^*(t) = \frac{1}{2M+1}\Bigg[&(2M+1) \\ &+ \left[\sum_{k=1}^{2M-1} a_k \cos((2\pi k f_m t + \gamma_k))\right] \\ &+ \cos((2\pi\cdot 2M\cdot f_m t + \phi_{-N} + \phi_N))\Bigg]. \end{aligned} \tag{A4}$$

Expression of a_k and γ_k are not necessary, but can be found analytically. The important point is that we have a pure phase modulation if

$$\begin{aligned} &\left[\sum_{k=1}^{2M-1} a_k \cos((2\pi k f_m t + \gamma_k))\right] \\ &\quad + \cos((2\pi\cdot 2M\cdot f_m t + \phi_{-N} + \phi_N)) = 0. \end{aligned} \tag{A5}$$

This expression cannot be verified because it would mean that the left part of Eq. (A5) is the Fourier series of the null function. Then, all of the coefficients of the Fourier series have to be null. However, according to Eq. (A5), the coefficient of the $2M \cdot f_m$ beat is equal to 1 in all the pure phase modulation cases. As a consequence, there is no possibility of finding a pure phase modulation that leads to a perfectly energy-balanced spectrum.

Appendix B

First, we consider a nonsinusoidal phase modulation with $N = 2$:

$$a_{\text{mod}}(t) = \exp(i[m_1 \sin(\omega_n t) + m_2 \sin(2\omega_n t + \varphi_2)]). \tag{B1}$$

We are looking for the values of φ_2 that lead to symmetric SPD. The general expression of SPD is $\text{SPD}(\omega) = |\tilde{a}_{\text{opt}}(\omega)|^2$. Thus, SPD is symmetrical if, for any value of ω, $\tilde{a}_{\text{mod}}(-\omega)$ is equal $e^{i\theta(\omega)}\tilde{a}_{\text{mod}}(\omega)$, with $\theta(\omega)$ as a real function (condition 1). According to Eq. (3), $\tilde{a}_{\text{mod}}(\omega)$ is obtained with Bessel functions as follows:

$$\tilde{a}_{\text{mod}}(\omega) = \left[\sum_n J_n(m_1)\delta(\omega - n\omega_n)\right] * \left[\sum_k J_k(m_2)e^{ik\varphi_2}\delta(\omega - 2k\omega_n)\right]. \tag{B2}$$

Equation (B2) can be rewritten as

$$\tilde{a}_{\text{mod}}(\omega) = \left[\sum_n \left[\sum_k J_{n-2k}(m_1) \cdot J_k(m_2)e^{ik\varphi_2}\right]\delta(\omega - n\omega_n)\right]. \tag{B3}$$

Using Bessel function propriety, $J_{-n}(m) = (-1)^n J_n(m)$, for any integer l, we have expression of each spectral peak:

$$\tilde{a}_{\text{mod}}(l\omega_n) = \left[\sum_k J_{l-2k}(m_1) \cdot J_k(m_2)e^{ik\varphi_2}\right] \quad \text{and}$$
$$\tilde{a}_{\text{mod}}(-l\omega_n) = (-1)^l\left[\sum_k J_{l-2k}(m_1) \cdot J_k(m_2)(-1)^k e^{-ik\varphi_2}\right]. \tag{B4}$$

Condition 1 is equivalent to the following expression for any integer l:

$$\tilde{a}_{\text{mod}}(-l\omega_n) = e^{i\theta(l\omega_n)}\tilde{a}_{\text{mod}}(l\omega_n). \tag{B5}$$

Using (B4), condition (B5) is satisfied for any m_1 and m_2 if, for any k,

$$(-1)^l(-1)^k e^{-ik\varphi_2} = e^{i\theta(l\omega_n)}e^{ik\varphi_2}. \tag{B6}$$

For $k = 0$, we have $e^{i\theta(l\omega_n)} = (-1)^l$, and so Eq. (B6) is simplified into $(-e^{-i\varphi_2})^k = (e^{-i\varphi_2})^k$. It is verified for any k if $-e^{-i\varphi_2} = e^{-i\varphi_2}$. Thus, $\cos(\varphi_2) = 0$, which means that $\varphi_2 = \frac{\pi}{2}[\pi]$.

For a symmetrical SPD, we have established the condition on φ_2 for any nonsinusoidal phase modulation with two harmonics: $\varphi_2 = \frac{\pi}{2}[\pi]$. The same kind of proof can be made with a three-harmonic phase-modulation signal. Thus, we consider the following phase-modulated signal:

$$\tilde{a}_{\text{mod}}(t) = \exp(i[m_1 \sin(\omega_n t) + m_2 \sin(2\omega_n t + \varphi_2) + m_3 \sin(3\omega_n t + \varphi_3)]). \tag{B7}$$

Analytically calculated spectrum $\tilde{a}_{\text{mod}}(\omega)$ leads to the following condition to get a symmetrical SPD:

$$(-1)^h(-e^{-i\varphi_2})^k(e^{-i\varphi_3})^l = e^{i\theta(h\omega_n)}(e^{i\varphi_2})^k(e^{-i\varphi_3})^l. \tag{B8}$$

For $k = l = 0$, we have $e^{i\theta(h\omega_n)} = (-1)^h$. Then, considering $l = 0$, we find a condition for the relative phases of the second harmonic: $\varphi_2 = \frac{\pi}{2}[\pi]$. $k = 0$ leads to $(e^{-i\varphi_3})^l = (e^{i\varphi_3})^l$, which is true for any l if $e^{-i\varphi_3} = e^{i\varphi_3}$. Thus, $\sin(\varphi_3) = 0$, which means $\varphi_3 = 0[\pi]$.

For a symmetrical SPD, we have established the conditions on φ_2 and φ_3 for any nonsinusoidal phase modulation with three harmonics: $\varphi_2 = \frac{\pi}{2}[\pi]$ and $\varphi_3 = 0[\pi]$.

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